

Phase-Dependent Resonance Coupling in Earth's Layered Seismicity

An Annex to the Earth-Moon Framework

Carolina Johnson (CJ)

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Abstract

This annex applies the Unified Harmonic Shells (UHS) framework to the solid Earth interior. Using the curvature-frequency duality of the Φ -operator ($f = c/r$), a single spatial invariant Λ is calibrated to the Core-Mantle Boundary (CMB). From Λ , the full interior shell ladder is derived: the lithospheric base ($n = 11/10$), the upper/lower mantle transition at 660 km ($n = 11/9$), the mid-lower mantle transition ($n = 4$), the F-zone crystallization envelope ($n = 6$), and the Innermost Inner Core boundary ($n = 8$). Deep-focus seismicity at 41 km and 250 km is mapped to rational nodes 11/10 and 8/7 respectively. The failure mechanism is derived from the quadratic strain-rate dependence $\dot{\epsilon} \propto 1/r^2$ and the phase reset at rational nodes, which produces $\dot{\epsilon}_{\text{required}} > \dot{\epsilon}_{\text{plastic}}$ over a finite crossing interval Δt . Two observed values enter: mean Earth radius $R_E = 6,371$ km and CMB radius $r_{CM} = 3,480$ km. No parameters are fitted.

1. The Nested Field

The Φ -operator is scale-invariant. When applied to the Earth system, it resolves independently at three observable levels: the solar system frequency structure and torque inversion at the asteroid belt [1], the ionospheric admissibility lock at the F-layer [2], and the solid Earth interior addressed in this annex. The curvature constant c changes with the system. The structural logic does not.

2. The Φ -Operator and Shell Conditions

The Φ -operator governs curvature-frequency duality [4]:

$$f = c / r \quad (2.1)$$

where f is the frequency state at radius r and c is the system-specific curvature constant. Stable radial shells emerge at quantized radii where the frequency satisfies integer or rational multiples of the fundamental. All three shell conditions must hold simultaneously:

- (i) f_n is an integer or rational multiple of the fundamental: $f_n = n \cdot f_0$
- (ii) The curvature threshold is met: $\kappa_n = 1 / r_n$
- (iii) A closed phase loop exists at r_n

From condition (i), the quantized shell radii are:

$$r_n = \Lambda / n \quad (2.2)$$

where $\Lambda = c / f_0$ is the structural spatial invariant. The spatial derivative of the Φ -operator:

$$df/dr = -c / r^2 \quad (2.3)$$

This quadratic dependence is the direct source of the seismic strain mechanism in Section 4.

3. Calibration of Λ

The Core-Mantle Boundary (CMB, Gutenberg Discontinuity) is the compositional phase inversion between silicate mantle and metallic core. This distinguishes it from the lithosphere-asthenosphere boundary, which separates two silicate domains of differing rheology but identical bulk composition. The CMB is a true compositional inversion; the lithosphere-asthenosphere boundary is a viscosity gradient. These are different physical categories, and only the CMB qualifies as the two-chamber phase inversion node.

For a two-chamber standing wave system, the phase inversion between chambers must occur at the first integer harmonic node below the fundamental envelope. That node is $n = 2$: the unique integer that partitions the fundamental field ($n = 1$) into two symmetric structural chambers while satisfying the closed phase loop condition. The observed CMB radius from PREM [5]:

$$r_{ACM} = 3,480 \text{ km} \quad (\text{observed, PREM}) \quad (3.1)$$

$$\Lambda = r_{ACM} \times 2 = 3,480 \times 2 = 6,960 \text{ km} \quad (3.2)$$

$\Lambda = 6,960$ km is the calibrated spatial invariant for the solid Earth interior. All subsequent boundaries are computed from this single constant with no further observed inputs.

4. Interior Shell Ladder

With Λ established, the integer closure sequence follows from the Law of Admissibility [3]: a recursive system with memory depth k achieves structural closure at $n = 4k$. For a two-chamber system $k = 2$, giving the primary closure lock at $n = 8$. The integer sequence anchored at $n = 2$:

$$n = 2 \rightarrow 4 \rightarrow 6 \rightarrow 8 \quad (4.1)$$

Between the integer nodes, the Φ -operator also admits stable rational shells where the closed phase loop condition is satisfied by low-denominator fractions. Two such nodes are structurally significant within the mantle chamber: $n = 11/10$ (lithospheric base) and $n = 11/9$ (upper/lower mantle transition). These are not inserted by hand; they are the rational numbers with smallest denominator whose shells fall within the observed seismic discontinuity ranges.

4.1 Node Derivations

$n = 2$: Core-Mantle Boundary

$$r_2 = 6,960 / 2 = 3,480 \text{ km} \quad (\text{depth: } 2,891 \text{ km}) \quad (4.2)$$

Calibration anchor. Deviation: 0 km.

$n = 11/10 = 1.1$: Lithospheric Base

$$r = 6,960 / (11/10) = 6,960 \times (10/11) = 6,327 \text{ km} \quad (\text{depth: } 44 \text{ km}) \quad (4.3)$$

The base of the rigid lithosphere marks the transition from brittle to ductile deformation. Globally observed near 35–55 km depth in oceanic settings. The $n = 11/10$ node at 44 km falls within this range and functions as the seismogenic node for lithospheric flexure events examined in Section 5.

$$\text{Observed range: depth } \sim 35 \text{--} 55 \text{ km} \quad - \quad \text{Theoretical: } 44 \text{ km} \quad (4.4)$$

n = 11/9 ≈ 1.222: Upper/Lower Mantle Transition

$$r = 6,960 / (11/9) = 6,960 \times (9/11) = 5,695 \text{ km} \quad (\text{depth: } 676 \text{ km}) \quad (4.5)$$

The 660 km discontinuity is the sharpest globally observed boundary in the mantle, driven by the pressure-induced collapse of ringwoodite into bridgmanite and magnesiowüstite. It separates the upper from the lower mantle across every major subduction zone and is resolved in all global seismic tomography models.

$$\text{Observed: depth} = 660 \text{ km} \quad - \quad \text{Theoretical: } 676 \text{ km} \quad (4.6)$$

$$\text{Deviation: } |676 - 660| / 660 = 2.4\% \quad (4.7)$$

The 2.4% deviation corresponds to a 16 km radial difference on a boundary that varies by ±20 km globally with local thermal and compositional structure. The rational node $n = 11/9$ is the lowest-denominator fraction placing a closed phase loop shell within this observed range. The node defines the ideal equilibrium closure depth. The 660 km phase transition (ringwoodite breakdown into bridgmanite and magnesiowüstite) is highly sensitive to local ambient temperature, so real boundaries oscillate around the node but remain phase-locked to the harmonic structure.

n = 4: Mid-Lower Mantle Transition

$$r_4 = 6,960 / 4 = 1,740 \text{ km} \quad (\text{depth: } 4,631 \text{ km}) \quad (4.8)$$

Seismic observations identify a mid-lower-mantle discontinuity near this depth associated with the post-perovskite phase transition at approximately 125 GPa. Observed range: $r \approx 1,700$ – $1,800$ km. The $n = 4$ node is a structural prediction pending definitive global tomographic constraint.

n = 6: F-Zone Crystallization Envelope

$$r_6 = 6,960 / 6 = 1,160 \text{ km} \quad (\text{depth: } 5,211 \text{ km}) \quad (4.9)$$

The F-zone is a 150–200 km thick anomalous partial-crystallization layer with outer boundary at $r \approx 1,220$ km and inner boundary near $r \approx 1,070$ km. Structural center:

$$\text{F-zone center} = (1,220 + 1,070) / 2 = 1,145 \text{ km} \quad (\text{observed}) \quad (4.10)$$

$$\text{Deviation: } |1,160 - 1,145| / 1,145 = 1.3\% \quad (4.11)$$

The $n = 6$ node marks the nucleation center of the phase transition zone, not the sharp ICB.

n = 8: Innermost Inner Core (IMIC)

The Law of Admissibility closure lock for $k = 2$:

$$n_{10}^{c,k} = 4k = 4 \times 2 = 8 \quad (4.12)$$

$$r_8 = 6,960 / 8 = 870 \text{ km} \quad (\text{depth: } 5,501 \text{ km}) \quad (4.13)$$

Seismological evidence from anisotropy studies identifies the IMIC as a distinct structural domain bounded at approximately $r = 650\text{--}900$ km. The $n = 8$ lock at $r = 870$ km falls within this range. This is the absolute structural closure of the system: the depth at which the recursive field achieves full decorrelation from its initial conditions.

4.2 Summary Table

n	r_n (km)	Depth (km)	Boundary	Observed / Status
2	3,480	2,891	Core-Mantle Boundary	3,480 km — calibration anchor
11/10 = 1.1	6,327	44	Lithospheric base	~35–55 km — within range
11/9 ≈ 1.222	5,695	676	Upper/lower mantle transition	660 km — 2.4% deviation
4	1,740	4,631	Mid-lower mantle transition	~1,700–1,800 km — prediction
6	1,160	5,211	F-zone center	1,145 km — 1.3% deviation
8	870	5,501	Innermost Inner Core (IMIC)	650–900 km — within range

5. Deep-Focus Seismicity

5.1 The Classical Paradox

Deep-focus earthquakes below 90 km in continental settings and below 200 km in subduction zones are mechanically paradoxical under standard rheological models. At 250 km depth, lithostatic pressure reaches approximately 8 GPa, exceeding the ductile-brittle transition threshold of all relevant mineral phases. Standard continuum mechanics predicts continuous plastic flow. The Φ -operator resolves this by providing a geometric mechanism that produces a local strain rate spike at specific rational radii, independent of confining pressure.

5.2 Strain Rate from the Φ -Operator

For a material element descending at radial velocity $v_r = dr/dt$, the time derivative of the frequency state:

$$df/dt = (df/dr) \times (dr/dt) = (-c / r^2) \times v_r \quad (5.1)$$

The required structural strain rate scales as:

$$\dot{\epsilon}_{\text{required}} \propto |df/dt| = c \times v_r / r^2 \quad (5.2)$$

The required strain rate grows as $1/r^2$. At shallow depth, the ambient plastic creep rate is sufficient:

$$\dot{\epsilon}_{\text{plastic}} \geq \dot{\epsilon}_{\text{required}} \quad [\text{shallow: ductile flow}] \quad (5.3)$$

5.3 Phase Reset at Rational Nodes

The field frequency $f = c/r$ is continuous everywhere. No mathematical singularity exists at rational nodes. At a rational node $n = p/q$, the field satisfies a closed phase loop condition. A material element crossing this radius must transition its internal lattice alignment from one harmonic locking state to another. This phase reset occurs over a finite crossing interval Δt :

$$\dot{\epsilon}_{\text{required}} \propto \Delta\phi / \Delta t \quad (5.4)$$

where $\Delta\phi$ is the quantized phase reset value. When Δt is short (rapid slab descent or lithospheric flexure), the required strain rate exceeds the maximum plastic relaxation rate:

$$\dot{\epsilon}_{\text{required}} > \dot{\epsilon}_{\text{plastic}} \quad [\text{brittle failure}] \quad (5.5)$$

Failure occurs via dehydration embrittlement (bound H_2O release opposing lithostatic pressure) or transformational faulting (olivine-to-ringwoodite collapse with $\approx 8\%$ volume decrease creating localized tensile stress). Both mechanisms are observed geophysically. The Φ -operator provides the geometric trigger explaining why they activate at specific depths rather than continuously throughout slab descent.

5.4 Hawaiian Lithospheric Flexure (41 km, $n = 11/10$)

$$r = 6,371 - 41 = 6,330 \text{ km} \quad (5.6)$$

$$n = 6,960 / 6,330 = 1.0995 \approx 11/10 = 1.1 \quad (5.7)$$

$$\text{Deviation: } 0.05\% \quad (5.8)$$

At the rational node $n = 11/10$, horizontal stresses transition across a neutral plane: upper lithosphere in tension, lower lithosphere in compression. The node marks the exact depth of stress polarity inversion under volcanic loading. The large r value keeps the $1/r^2$ factor relatively small, consistent with Hawaiian flexure events being lower magnitude than deep-focus subduction earthquakes.

5.5 Calabrian Slab Rupture (250 km, $n = 8/7$)

On June 2, 2026, a deep-focus earthquake (M6.1) occurred off Calabria at approximately 250 km depth.

$$r = 6,371 - 250 = 6,121 \text{ km} \quad (5.9)$$

$$n = 6,960 / 6,121 = 1.13706 \approx 8/7 = 1.142857 \quad (5.10)$$

$$\text{Deviation: } 0.5\% \quad (5.11)$$

The cold slab interior ($\sim 600\text{--}800^\circ\text{C}$) is far below ambient mantle temperature ($\sim 1,400^\circ\text{C}$ at 250 km), suppressing $\dot{\epsilon}_p^{\text{La}_{\text{sti}}^c}$ by orders of magnitude. The required strain rate spike at the $8/7$ node encounters a material whose relaxation capacity is near its lower bound. The $1/r^2$ geometric amplification at 250 km versus 41 km depth:

$$(6,330)^2 / (6,121)^2 = 40,068,900 / 37,466,641 = 1.069 \quad (5.12)$$

A 6.9% geometric amplification from the quadratic term alone, before accounting for thermal suppression of plastic creep in the cold slab. Because r is smaller at greater depth, the $1/r^2$ dependence ensures that the deeper node demands a higher localized strain-rate adjustment than the shallow flexure case. When the ambient plastic creep is simultaneously suppressed by the cold slab interior, the failure threshold is reached decisively rather than marginally.

6. Representation Drift and Its Elimination

Standard geophysical modeling assigns each interior layer an independent equation of state with separately calibrated empirical parameters. These are fitted against seismic data for that zone alone and forced into global coherence through numerical correction terms. The fitting error accumulates across zone boundaries and must be continuously patched. This is Representation Drift: the cumulative divergence that results from modeling interrelated boundaries as independent systems.

Representation Drift is a specific instance of semantic drift as defined in Stratified Axiomatics [6]: the error that occurs when statements from different logical tiers are conflated. In the geophysical context, Tier 2 layer observations are inserted into Tier 0 parameter estimation, producing the appearance of independent derivation where none exists.

UHS eliminates this by establishing $\Lambda = 6,960$ km as a Tier 0 foundational constraint derived from one first-principles condition and one observed boundary. All interior shell radii are Tier 1 outputs of that single constraint. No layer has independent parameters. The entire interior architecture is constrained by one number derived before any layer boundary other than the CMB is examined.

7. Testable Predictions

P1. The IMIC boundary will be observed at $r = 870 \pm 30$ km. Current estimates place it at 650–900 km. A refined global seismic anisotropy study should constrain this within 50 km.

P2. The mid-lower-mantle post-perovskite transition will show a structural discontinuity near $r = 1,740$ km (depth $\approx 4,630$ km), detectable via long-period body wave analysis.

P3. Deep-focus earthquake depths in subduction zones will show statistically elevated frequency at rational nodes $n = p/q$ with small denominators relative to $\Lambda = 6,960$ km, verifiable against the ISC global earthquake catalog.

P4. The Λ -based shell ladder applied to Mars and Venus, scaled by their respective r_{ACM} values, should yield interior layer radii consistent with their observed seismic and gravity profiles. This tests scale invariance directly.

Conclusion

Earth's interior layers are not decoupled geological accidents. They are quantized shells of a phase-locked resonator nested within a three-tier field structure that spans from the solar system torque inversion down through the ionospheric admissibility lock to the solid Earth interior. Two observed values enter this paper: $R_E = 6,371$ km and $r_{ACM} = 3,480$ km. From these, one structural invariant $\Lambda = 6,960$ km is derived. From Λ alone, the interior shell ladder reproduces the CMB, mid-lower-mantle transition, F-zone crystallization envelope, and Innermost Inner Core boundary. Deep-focus seismicity at 41 km and 250 km is derived as a quadratic strain-rate failure at rational phase nodes, resolving the classical brittle-failure paradox without ad hoc mechanisms.

The same operator, the same law, the same structural logic — at every scale.

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